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Ans 1. Newton's second law: rate of change of momentum is proportional to applied force, direction of force. $F = ma$.

Ans 2. Speed is scalar (magnitude only), e.g. 40 km/hr.

Velocity is vector (magnitude + direction), e.g. 40 km/hr north.

Ans 4. SI unit of current: Ampere. One ampere = one coulomb per second.

Ans 6. $u=0$, $a=2$ m/s^2, $t=5$s. $v = u+at = 10$ m/s.

$s = ut + 1/2at^2 = 25$ m.

Ans 8. Ohm's law: $V=IR$. $V = 2 \times 10 = 20$ Volts.

Ans 11 (a). $v=u+at$ ... average velocity$=(u+v)/2$, s = avg x t

substituting: $s = ut + 1/2at^2$

Ans 9. Three states: solid, liquid, gas — particle spacing differs.

Ans 7.

Mitosis: 2 identical cells (growth). Meiosis: 4 cells, half chromosomes (reproduction).

Ans 12. Rainwater harvesting; fixing leaks promptly.

formula sheet: $F=ma$, $v=u+at$, $s=ut+1/2at2$, $V=IR$